HW #1

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. For an R-algebra S, we say that S is **integral** over R if every element $s \in S$ is integral over R.

Suppose that S is an R-algebra and T is an S-algebra. Show that if S is integral as an R-algebra and if T is integral as an S-algebra, then T is integral as an R-algebra.

Proof. We have to show that, for any $t \in T$, t is integral over R. As t is integral over S, there is a monic polynomial $p(X) \in S[X]$ such that $p(t) = 0$. Let $p(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$. Consider $S' := R[a_0, \cdots, a_{n-1}]$; namely, the R-subalgebra of S generated by $a_0, \cdots, a_{n-1}$. Then, as $t^n = -(a_{n-1}t^{n-1} + \cdots + a_0)$, t is integral over S', or equivalently $S'[t]$ (the S'-subalgebra of T generated by t) is finitely generated as an S'-module.

On the other hand, as each of $a_0, \cdots, a_{n-1}$ is integral over R, it follows that S' is finitely generated as an R-module. Therefore, $S'[t]$, being a finitely generated S'-module, is also finitely generated as an R-module. Obviously, $S'[t]$ is an R-algebra, as it is an S'-algebra. By Theorem A.11(1) of the textbook, therefore, this implies that t is integral over R. $\square$

Question 2. Let K be a number field, and L be a subfield. For $a \in \mathcal{O}_K$, let $p_a(X) \in L[X]$ be the minimal polynomial of a over L. Show that $p_a(X) \in \mathcal{O}_L[X]$.

Proof. Let F be the Galois closure of K over L. Then, $p_a(X) = (X - \sigma_1(a)) \cdots (X - \sigma_d(a))$ for $\sigma_1, \cdots, \sigma_d \in \mathrm{Gal}(F/L)$. As a is an algebraic integer, $\sigma_i(a)$ is also an algebraic integer for any $1 \leq i \leq d$. Therefore, the coefficients of $p_a(X)$ are $\mathbb{Z}$-linear combinations of products of $\sigma_i(a)$'s, which are all algebraic integers. Therefore, the coefficients of $p_a(X)$ are algebraic integers. As $p_a(X) \in L[X]$, this implies that $p_a(X) \in \mathcal{O}_L[X]$. $\square$

Question 3. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which is a number field with $[K : \mathbb{Q}] = 4$. Show that

$$\mathcal{O}_K = \left\{ a + \frac{b\sqrt{2}}{2} + c\sqrt{3} + \frac{d\sqrt{6}}{2} \ : \ a, b, c, d \in \mathbb{Z}, b \equiv d \pmod 2 \right\}.$$

Hint. One way to simplify the calculation is to use Question 2, using the subfields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ of K.

Proof. We use **Hint**. An element of K is uniquely expressed as $w + x\sqrt{2} + y\sqrt{3} + z\sqrt{6}$, $w, x, y, z \in \mathbb{Q}$ (i.e., $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is a $\mathbb{Q}$-basis of K as a $\mathbb{Q}$-vector space). Suppose that $\alpha = w + x\sqrt{2} + y\sqrt{3} + z\sqrt{6} \in \mathcal{O}_K$. Then, its minimal polynomial over $\mathbb{Q}(\sqrt{2})$ is given by

$$X^2 - \mathrm{Tr}_{K/\mathbb{Q}(\sqrt{2})}(\alpha)X + N_{K/\mathbb{Q}(\sqrt{2})}(\alpha) \in \mathcal{O}_{\mathbb{Q}(\sqrt{2})}[X].$$

Therefore, $\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{2})}(\alpha), N_{K/\mathbb{Q}(\sqrt{2})}(\alpha) \in \mathcal{O}_{\mathbb{Q}(\sqrt{2})} = \mathbb{Z}[\sqrt{2}]$ (we calculated $\mathcal{O}_{\mathbb{Q}(\sqrt{2})}$ in class). Note that

$$\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{2})}(\alpha) = \alpha + \sigma(\alpha), \quad N_{K/\mathbb{Q}(\sqrt{2})}(\alpha) = \alpha\sigma(\alpha),$$

where $\sigma : K \to K$ is the nontrivial element in $\mathrm{Gal}(K/\mathbb{Q}(\sqrt{2}))$, which sends $\sqrt{2} \mapsto \sqrt{2}$ and $\sqrt{3} \mapsto -\sqrt{3}$. Thus

$$\sigma(\alpha) = w + x\sqrt{2} - y\sqrt{3} - z\sqrt{6}$$

$$\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{2})}(\alpha) = (w + x\sqrt{2} + y\sqrt{3} + z\sqrt{6}) + (w + x\sqrt{2} - y\sqrt{3} - z\sqrt{6}) = 2w + 2x\sqrt{2}$$

$$\begin{aligned} N_{K/\mathbb{Q}(\sqrt{2})}(\alpha) &= (w + x\sqrt{2} + y\sqrt{3} + z\sqrt{6})(w + x\sqrt{2} - y\sqrt{3} - z\sqrt{6}) \\ &= (w + x\sqrt{2})^2 - 3(y + z\sqrt{2})^2 \\ &= \left((w^2 + 2x^2) + 2wx\sqrt{2}\right) - 3\left((y^2 + 2z^2) + 2yz\sqrt{2}\right) \\ &= (w^2 + 2x^2 - 3y^2 - 6z^2) + \sqrt{2}(2wx - 6yz). \end{aligned}$$

Similarly, we can do this with $K/\mathbb{Q}(\sqrt{3})$, and use that $\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{3})}(\alpha), N_{K/\mathbb{Q}(\sqrt{3})}(\alpha) \in \mathcal{O}_{\mathbb{Q}(\sqrt{3})} = \mathbb{Z}[\sqrt{3}]$. Note that

$$\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{3})}(\alpha) = 2w + 2y\sqrt{3}$$

$$\begin{aligned} N_{K/\mathbb{Q}(\sqrt{3})}(\alpha) &= (w + y\sqrt{3})^2 - 2(x + z\sqrt{3})^2 \\ &= \left((w^2 + 3y^2) + 2wy\sqrt{3}\right) - 2\left((x^2 + 3z^2) + 2xz\sqrt{3}\right) \\ &= (w^2 - 2x^2 + 3y^2 - 6z^2) + \sqrt{3}(2wy - 4xz). \end{aligned}$$

From $\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{2})}(\alpha) \in \mathbb{Z}[\sqrt{2}]$, we get that $w, x \in \frac{1}{2}\mathbb{Z}$, and similarly from $\mathrm{Tr}_{K/\mathbb{Q}(\sqrt{3})}(\alpha) \in \mathbb{Z}[\sqrt{3}]$, we get that $w, y \in \frac{1}{2}\mathbb{Z}$. Let $w = \frac{s}{2}$, $x = \frac{t}{2}$, $y = \frac{u}{2}$ for $s, t, u \in \mathbb{Z}$. Then, the norms being algebraic integers become the conditions

$$(1) \quad \cdots\cdots \quad \frac{s^2 + 2t^2 - 3u^2 - 24z^2}{4} \in \mathbb{Z}$$

$$(2) \quad \cdots\cdots \quad \frac{st - 6uz}{2} \in \mathbb{Z}$$

$$(3) \quad \cdots\cdots \quad \frac{s^2 - 2t^2 + 3u^2 - 24z^2}{4} \in \mathbb{Z}$$

$$(4) \quad \cdots\cdots \quad \frac{su - 4tz}{2} \in \mathbb{Z}$$

Recall that so far $s, t, u \in \mathbb{Z}$ and $z \in \mathbb{Q}$. It follows from (1) that $24z^2 \in \mathbb{Z}$, which implies that $z = \frac{v}{2}$ for $v \in \mathbb{Z}$. Replacing z with $v/2$, the conditions become

$$(1) \quad \cdots\cdots \quad \frac{s^2 + 2t^2 - 3u^2 - 6v^2}{4} \in \mathbb{Z}$$

$$(2) \quad \cdots\cdots \quad \frac{st - 3uv}{2} \in \mathbb{Z}$$

$$(3) \quad \cdots\cdots \quad \frac{s^2 - 2t^2 + 3u^2 - 6v^2}{4} \in \mathbb{Z}$$

$$(4) \quad \cdots\cdots \quad \frac{su - 2tv}{2} \in \mathbb{Z}$$

where now $s, t, u, v \in \mathbb{Z}$. By (1), $s^2 - 3u^2$ is even; therefore, $s \equiv u \pmod 2$. On the other hand, by (4), su is even. Therefore, both s, u are even. Therefore, $s = 2a$ and $u = 2c$ for some $a, c \in \mathbb{Z}$. Replacing these, we see that the conditions now become

$$(1) \quad \cdots\cdots \quad \frac{s^2 + 2t^2 - 3u^2 - 6v^2}{4} \in \mathbb{Z} \quad \rightsquigarrow \quad t^2 \equiv v^2 \pmod 2$$

$$(2) \quad \cdots\cdots \quad \frac{st - 3uv}{2} \in \mathbb{Z} \quad \rightsquigarrow \quad \text{vacuous}$$

$$(3) \quad \cdots\cdots \quad \frac{s^2 - 2t^2 + 3u^2 - 6v^2}{4} \in \mathbb{Z} \quad \rightsquigarrow \quad t^2 \equiv v^2 \pmod 2$$

$$(4) \quad \cdots\cdots \quad \frac{su - 2tv}{2} \in \mathbb{Z} \quad \rightsquigarrow \quad \text{vacuous}$$

Therefore, the remaining conditions imply that $t \equiv v \pmod 2$. Setting $b = t$ and $d = v$, we get the desired result. $\square$